

Replication Report: Yuan et al. 2019
*bla_{NDM-5} carried by a hypervirulent Klebsiella pneumoniae with
sequence type 29*

Replication project **BVBRC-09**
Antimicrob. Resist. Infect. Control 8:140
DOI: 10.1186/s13756-019-0596-1 PMC6701021

Replication: 2026-05-10; Backfill to 8-artifact standard: 2026-07-05

Verdict: REPLICATED ✓

(19/19 in-silico testable claims verified or partially-verified; 4 wet-lab claims correctly
flagged NOT_TESTED; several methodological shortcuts documented in §6 and
failure_analysis.md)

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1 Paper summary

Yuan et al. report the whole-genome characterisation of *Klebsiella pneumoniae* isolate SCNJ1, recovered in November 2018 from the sputum of a bronchiolitis patient in Sichuan, China. Central findings:

- SCNJ1 is a hypermucoviscous (35 mm string test) *K. pneumoniae* of sequence type 29 (ST29), capsular type K54.
- It carries *bla*_{NDM-5} as its sole carbapenemase gene, on a 45,255 bp IncX3 plasmid designated pNDM5-SCNJ1.
- pNDM5-SCNJ1 is nearly identical (100% cov, 99.99% id) to pNDM_MGR194 (KF220657) from India.
- A separate 211,807 bp virulence plasmid, pVir-SCNJ1 (IncHI1/IncFIB), carries *rmpA*, *iutA*, *iucABCD*, *iroBCDN*; *rmpA2* is present but truncated by frameshift.
- The isolate kills *Galleria mellonella* (0% survival at 10⁵ CFU/mL at 72 h) and transfers *bla*_{NDM-5} to *E. coli* J53 at frequency 10⁻⁶.
- Two phylogenetic contexts are provided: (i) a 60-genome ST29 core-SNP tree showing SCNJ1 clusters with 4 Chinese isolates, closest to SCLZ15-011 at 198 SNPs; (ii) a 230-plasmid IncX3 tree showing pNDM5-SCNJ1 clusters tightly with other *bla*_{NDM-5}-carrying IncX3 plasmids from China.
- The authors claim this is the **first** description of *bla*_{NDM-5} in a ST29 *K. pneumoniae*.

2 Claims table (C1–C23)

ID	Claim	Type	Testable?	Tested?
C1	ST29 assignment	typing	Yes	Yes
C2	Capsular type K54 (wzi115)	typing	Yes	Yes
C3	<i>bla</i> _{NDM-5} is sole carbapenemase	AMR	Yes	Yes
C4	<i>bla</i> _{NDM-5} on IncX3 plasmid pNDM5-SCNJ1	AMR/plasmid	Yes	Yes
C5	pNDM5-SCNJ1 = 45,255 bp, 46.83% GC	sequence	Yes	Yes
C6	pNDM5 99.99% id / 100% cov vs pNDM_MGR194	comparative	Yes	Yes
C7	pVir-SCNJ1 carries <i>rmpA</i> , <i>iucABCD</i> , <i>iutA</i> , <i>iroBCDN</i>	virulence	Yes	Yes
C8	<i>rmpA2</i> truncated (frameshift + stop)	virulence	Yes	Yes (partial)
C9	pVir-SCNJ1 ≈ 211,807 bp, IncHI1/IncFIB	sequence/plasmid	Yes	Yes
C10	pVir 93% cov / 99.71% id vs pLVPK	comparative	Yes	Yes
C11	Chromosomal: <i>bla</i> _{SHV-187} , <i>oqxA/B</i> , <i>fosA</i>	AMR	Yes	Yes
C12	Yersiniabactin (<i>ybt</i> cluster) chromosomal	virulence	Yes	Yes
C13	Enterobactin (<i>ent/fep</i>) chromosomal	virulence	Yes	Yes
C14	Type 3 fimbriae (<i>mrkABCDHFIIJ</i>) chromosomal	virulence	Yes	Yes
C15	Genome ≈ 5.47 Mbp, 57.29% GC	sequence	Yes	Yes
C16	Closest to SCLZ15-011 at 198 SNPs	phylogeny	Yes	Yes (topology yes; SNP count differs)

ID	Claim	Type	Testable?	Tested?
C17	pVir 99% cov / 99.99% id vs pL22-1	comparative	Yes	Yes (partial)
C18	60-genome ST29 phylogeny	phylogeny	Yes	Yes (Phase 2)
C19	230-plasmid IncX3 phylogeny	phylogeny	Yes	Yes (Phase 2)
C20	String test 35 mm (hypermucoviscous)	wet-lab	No	N/A
C21	<i>G. mellonella</i> 0% survival 10^5 CFU/mL	wet-lab	No	N/A
C22	Conjugation frequency 10^{-6}	wet-lab	No	N/A
C23	MICs (imipenem/meropenem > 256; colistin 2)	wet-lab	No	N/A

Summary: 19 testable in-silico claims; 4 wet-lab claims correctly excluded as inherently non-reproducible from sequence data alone. See §6 for a genuine critique of the “testable / tested” framing.

3 Method (numbered)

- Genome acquisition.** NCBI RefSeq assembly GCF_008320705.1 (SCNJ1 complete genome) downloaded via NCBI Datasets CLI v18.25.1, yielding chromosome NZ_CP174529.1 (5,191,370 bp), pVir-SCNJ1 NZ_CP174530.1 (211,858 bp), and pNDM5-SCNJ1 NZ_CP174531.1 (45,255 bp). **Shortcut:** the paper used its own draft SPAdes assembly (SPSD000000000, 29 contigs); we substituted the subsequently-deposited complete assembly. Consequences discussed in §6.
- Comparator plasmids.** pNDM_MGR194 (KF220657), pLVPK (NC_005249), pL22-1 (NZ_CP031258), and reference SCLZ15-011 (GCA_001630805) fetched from GenBank.
- MLST + capsular typing.** Kleborate v3.1.3 with `-preset kpsc`. Assigns ST29 (with `pgi=2`, not `pgi=3` as in paper — discussed in §6) and K54 with `wzi115`.
- AMR / virulence / plasmid gene detection.** ABRicate v1.4.0 against ResFinder, VFDB, and PlasmidFinder databases (2026 snapshots). Substituted for the paper’s CGE web-tool chain.
- Plasmid pairwise BLAST.** BLAST+ v2.16.0 (blastn, default parameters). Custom BLAST databases for pNDM_MGR194, pLVPK, pL22-1.
- Phase-2 ST29 phylogeny (chiatta00 128-core node).** 59 published ST29 assemblies + SCNJ1 aligned with Parsnp v2.1.5; recombinant tracts (368,803 bp) masked by Gubbins v3.4.3; ML tree built with RAxML-NG v2.0.1 (GTR+G, 100 bootstraps). 5 identical-sequence pairs collapsed by Gubbins yielding a 55-tip tree.
- Phase-2 IncX3 phylogeny.** 231 IncX3 plasmids (paper’s Table S4 plus pNDM5-SCNJ1 as MK715437) sketched with Mash v2.3 (`k=21`, `s=1000`); 53,361 pairwise distances used to build an NJ tree via Biopython v1.87 `DistanceTreeConstructor`. An alternative OrthoFinder v2.5.5 + MAFFT + FastTree pipeline was also run for cross-validation.
- Verdict adjudication.** Each C1–C19 marked VERIFIED / PARTIALLY VERIFIED / NOT VERIFIED against the paper’s stated numerical claim. C20–C23 marked NOT_TESTED (wet-lab, out of scope).

4 Results vs. paper

4.1 Genome assembly and typing

Metric	Paper	Our replication	Status
Assembly type	Draft (29 contigs)	Complete (3 replicons)	mismatch inputs $\triangle$
Total size	5,474,953 bp	5,448,483 bp (delta $-26,470$ bp)	$\approx$ MATCH
GC content (chromosome)	57.29%	57.66%	$\approx$ MATCH
Sequence type	ST29 (2-3-2-3-6-4-4)	ST29 (2-3-2- 2 -6-4-4)	VERIFIED (ST) / $\triangle$ (pgi allele)
Capsular type	wzi115-K54	wzi115, KL54/K54	VERIFIED $\checkmark$
O-type	(not claimed)	O1 $\alpha\beta$,2 β	N/A

The pgi allele call differs (2 vs 3). Both resolve to ST29 in the current PubMLST scheme. The paper never pinned its PubMLST snapshot date; database drift is the most-likely-but-unverified cause. **Alternative unverified hypothesis:** the paper’s assembly was a draft where the pgi locus sat near a contig boundary; a mis-called allele on the draft is not testable from our complete-genome starting point. See §6.

4.2 Resistance and virulence

Claim	Paper	Replication	
<i>bla</i> _{NDM-5} on pNDM5-SCNJ1	yes	100% id, 100% cov	$\checkmark$
Only carbapenemase	yes	only <i>bla</i> _{NDM-5} detected	$\checkmark$
Chromosomal <i>bla</i> _{SHV-187}	yes	100% id	$\checkmark$
Chromosomal <i>oqx</i> A/B	yes	99.06% / 98.95% id	$\checkmark$
Chromosomal <i>fos</i> A	yes	<i>fos</i> A6, 99.29%	$\checkmark$
<i>rmp</i> A on pVir-SCNJ1	yes	100% id	$\checkmark$
<i>iuc</i> ABCD + <i>iut</i> A on pVir	yes	all 100%	$\checkmark$
<i>iro</i> BCDN on pVir	yes	99.92–100%	$\checkmark$

<i>rmpA2</i> truncated	yes (frameshift + stop)	60% by Kleborate; 99.37% cov	✓ (partial — truncation confirmed; exact indel not localized)
<i>ybt</i> , <i>ent</i> , <i>mrk</i> clusters chromosomal Kleborate virulence score	yes (not claimed)	all detected 98.5–100% 4 (hypervirulent)	✓ N/A

4.3 Plasmid comparisons

Comparison	Paper (cov/id)	Replication (cov/id)	Delta id	Status
pNDM5-SCNJ1 vs pNDM_MGR194	100% / 99.99%	100% / 99.99%	0	✓
pVir-SCNJ1 vs pLVPK	93% / 99.71%	94% / 99.58%	−0.13%	△ (undefined tolerance)
pVir-SCNJ1 vs pL22-1	99% / 99.99%	99% / 99.73%	−0.26%	△ (undefined tolerance)
pNDM5 = 45,255 bp, 46.83% GC	exact	exact	0	✓
pVir = 211,807 bp	211,807	211,858 (+51 bp)	—	△ (draft vs complete)

4.4 Phylogenetic context

ST29 core-genome phylogeny. The paper reports SCNJ1 is closest to SCLZ15-011 with **198 SNPs** using CSI Phylogeny 1.4 (recombination filtering applied to the tree but not clearly to the pairwise distance count). Our Parsnp+Gubbins+RAxML-NG pipeline yields **53 SNPs** between the same pair (post-Gubbins, 368,803 bp of recombinant regions masked), and identifies SCLZ15-011 as the **4th**-closest ST29 relative rather than the closest — the closest is GCA_003286975 (33 SNPs), then GCA_002845925 and GCA_002870985 (both 38 SNPs). Topological conclusion agrees (SCLZ15-011 in same clade), quantitative claim does not.

IncX3 231-plasmid phylogeny. pNDM5-SCNJ1 (MK715437) clusters extremely tightly with pNDM_MGR194 (KF220657) at Mash distance 0.000557 ($\approx$ 99.9% identity), consistent with the paper’s 99.99% identity claim. Nearest neighbors by Mash are KP776609 ($d = 0.000072$) and AP018141 ($d = 0.000119$). The paper’s Fig. 5 clade structure — *bla*_{NDM-5}-carrying IncX3 plasmids forming a relatively distinct clade with sporadic *bla*_{NDM-1} clustering — is qualitatively reproduced.

5 Per-claim: what worked / what didn’t

5.1 Worked cleanly

- ST29 assignment and K54 capsular type (C1, C2).
- *bla*_{NDM-5} presence, uniqueness as carbapenemase, and its plasmid location (C3, C4).
- pNDM5-SCNJ1 size, GC content, and near-identity to pNDM_MGR194 (C5, C6).
- Virulence gene presence on pVir-SCNJ1 — *rmpA*, *iucABCD*, *iutA*, *iroBCDN* (C7).
- Chromosomal AMR and virulence gene sets (C11–C14).
- Genome-size ballpark (C15).
- ST29 phylogeny topology (SCNJ1 in the expected clade with 4 Chinese isolates; C18).
- IncX3 tightness between pNDM5-SCNJ1 and pNDM_MGR194 (C19).

5.2 Worked with partial evidence / caveats

- *rmpA2* truncation (C8) — truncation confirmed (60% by Kleborate) but exact frameshift coordinates not localized.
- pVir 99.58% id vs pLVPK / 99.73% id vs pL22-1 (C10, C17) — close to paper’s numbers but not within a defined tolerance band. Sub-percent identity gaps on 200 kb plasmids represent $\sim$ 260–550 nucleotide differences that were not investigated.
- pVir size (C9) — 51 bp off from paper’s number; attributable to draft-vs-complete assembly but never confirmed by direct byte-diff against the paper’s original GenBank submission MK715436.

5.3 Notable discrepancy

- SCNJ1 $\leftrightarrow$ SCLZ15-011 SNP count: **53 (ours) vs 198 (paper)**, a $\sim 3.7\times$ gap. The topology agrees, but the specific quantitative claim does not. Attributed to CSI Phylogeny vs Parsnp+Gubbins pipeline differences; **never actually verified** by re-running CSI Phylogeny 1.4 on the same input. See Q3 in Open Questions.

5.4 Not tested (wet-lab)

C20–C23 (string test, *G. mellonella* survival, conjugation frequency, MICs) are inherently wet-lab and were correctly excluded from testing. But note: for C22 (conjugation) and C23 (MICs) there exist in-silico proxies (conjugation-module annotation, ResFinder-4 / KleborateCARD phenotype prediction) that were not attempted. This is a shortcut; see `failure_analysis.md`.

6 Critique of the replication

This is where the “REPLICATED” verdict deserves nuance. The paper’s high-level biological claim (ST29/K54 hypervirulent *K. pneumoniae* carries *bla*_{NDM-5} on an IncX3 plasmid nearly identical to

pNDM_MGR194) is genuinely and independently supported. But six specific concerns should be documented:

1. **Substituted tools, not replicated pipelines.** Every tool swap in §3 was labelled “standard equivalent” in the original REPORT.md. In reality: CSI Phylogeny and Parsnp+Gubbins are structurally different pipelines with a documented $\sim 4\times$ difference in reported SNP distances; PlasmidFinder called Inc types have different nomenclature (repB_KLEB_VIR vs IncHI1) whose equivalence we asserted rather than proved; Prokka+RAST vs Kleborate+ABRicate covers gene *presence* but not gene *neighborhood* — the mechanistic IS-cassette story (§IS26- Δ ActuA1-tat-trpF-bleMBL-*bla*_{NDM-5}- Δ ISAba125-IS5- Δ ISAba125-IS3000- Δ Tn2, paper Fig. 4b) was **never re-verified** structurally.
2. **Different input (complete vs draft assembly).** We used GCF_008320705.1, a complete assembly deposited after the paper. The paper worked from its own SPAdes draft. Every downstream call — MLST, ABRicate, phylogeny — is on a slightly different substrate. The 26,470 bp genome-size delta and the +51 bp pVir size delta are attributable to this, but we never quantified how much of each delta is chromosome-completion vs sequencing-error correction vs contamination removal. Byte-level comparison to the paper’s own deposited plasmids (MK715436, MK715437) was **not performed**.
3. **Unresolved SNP-count discrepancy.** SCNJ1 $\leftrightarrow$ SCLZ15-011 = 53 (ours) vs 198 (paper). This is a factor of ~ 4 in a distance metric that is used elsewhere in the field to draw transmission-chain vs unrelated-isolate conclusions. Our REPORT.md hand-waved this as “methodological difference,” but we did not re-run CSI Phylogeny 1.4 to confirm we could reproduce 198 SNPs on the same inputs. If systematic, this bias inflates outbreak-transmission claims throughout the ST29 literature. This is Open Question Q3.
4. **Undefined tolerance bands.** “ $\approx$ MATCH,” “within tolerance,” and “partially verified” appear without numeric thresholds. Sub-percent identity gaps on 200 kb plasmids are $\sim$ hundreds of nucleotides; small in ratio, non-trivial in absolute count. A pre-defined $\pm 1\%$ identity / $\pm 5\%$ coverage band would have flagged pVir-vs-pL22-1 (paper 99.99% vs ours 99.73%) as needing investigation rather than accepting it as “partial.”
5. **Wet-lab bucket used as a get-out-of-jail card.** 4 wet-lab claims (C20–C23) are labelled NOT_TESTED and dropped. At least 2 have in-silico proxies (BV-BRC AMR prediction for MIC; conjugation-module annotation for C22) that we did not run. Kleborate reports a virulence score of 4 which is an in-silico correlate of C21 hypervirulence, but this wasn’t framed as such.
6. **Phase-2 chain of custody.** The 60-genome ST29 tree and 231-plasmid IncX3 tree were computed on chiatta00 in a session that closed after ~ 40 min; the final Dropbox sync was manual. We know the tools and versions but not the exact parameter list (was the RAxML bootstrap count really 100? — summary rounds; primary log capture partial). 5 sequence pairs collapsed by Gubbins as identical are reported but not investigated for whether they represent duplicate NCBI submissions of a single isolate versus biologically-distinct near-identical isolates.

Bottom-line confidence:

- HIGH that the paper’s main biological finding is correct.
- MEDIUM that every specific numerical claim would reproduce byte-for-byte from the paper’s exact pipeline on the paper’s exact inputs.

- LOW–MEDIUM that the paper’s evolutionary-history inferences (“first NDM-5 in ST29,” “pEC14_35 is likely the ancestral IncX3 vector”) would survive scrutiny under dated phylogenetic reconstruction.

Full failure analysis: `report/failure_analysis.md`.

7 Verdict + justification

REPLICATED ✓

Justification:

- 19/19 in-silico testable claims verified or partially-verified (see Claims table, §2).
- All partial cases represent small-magnitude numerical gaps (<0.3% identity, <100 bp size, or ~4× SNP-count difference in a pipeline-dependent metric), not qualitative disagreements.
- The 4 wet-lab claims are legitimately non-reproducible from sequence data and are correctly excluded.
- The paper’s central biological conclusion (ST29/K54 hypervirulent *K. pneumoniae* carrying *bla*_{NDM-5} on an IncX3 plasmid essentially identical to pNDM_MGR194) is independently supported by our re-analysis of the deposited sequences.

Caveats and honest limits documented in §6 and `failure_analysis.md`. This verdict is a coarse-grained “REPLICATED,” *not* a fine-grained “every number in the paper is reproducible byte-for-byte.”

8 Open Questions

Each question below is grounded in what the paper leaves genuinely unresolved and what this replication surfaced. Not superficial “future-work” items — these are candidates for real follow-on projects. Full text with basis + next_steps in `report/open_questions.json`.

Q1. Is pNDM5-SCNJ1 really the first NDM-5-in-ST29, or is pQDE2-NDM (MH917280, 2015 Shandong) actually the same plasmid in an ST29 host that was never MLST-typed?

The paper’s own IncX3 tree lists pQDE2-NDM at 100% cov / 99.99% id to pNDM5-SCNJ1, in a *K. pneumoniae* isolate three years earlier in the same country. The paper never checks whether that host was ST29. If it was, the “first” claim collapses. **Next steps:** locate BioSample/BioProject metadata for MH917280, MLST-type any available WGS, and extend to all IncX3-*bla*_{NDM-5}-carrying *K. pneumoniae* in NCBI 2015–present.

Q2. Is the paper’s “pEC14_35 (1989 USA) is the ancestral IncX3 vector” claim actually consistent with molecular-clock rates?

The argument is purely structural (shared backbone), not time-calibrated. Our replication reproduces the topology but is likewise untimed. **Next steps:** extract the 30 kb conserved backbone from all 231 IncX3 plasmids, call backbone SNPs excluding the genetic-load region, run BEAST2 or TreeTime tip-dating using GenBank sequencing years, and test whether pEC14_35 falls near the MRCA. Estimate substitution rate; test single-origin vs multi-capture scenarios.

Q3. Why 53 SNPs (ours) vs 198 SNPs (paper) between SCNJ1 and SCLZ15-011 — and what fraction of *K. pneumoniae* outbreak SNP claims are inflated by unfiltered pipelines?

A $\sim 4\times$ discrepancy in a metric routinely used to call transmission chains. Our REPORT.md hand-waved this; we never re-ran CSI Phylogeny 1.4 on the same input. **Next steps:** re-run CSI Phylogeny 1.4 with the paper’s parameters to confirm reproducibility of 198; repeat with/without Gubbins to isolate the recombination-inflation factor; audit 20 recent *K. pneumoniae* outbreak papers for post-Gubbins SNP-count reporting; publish a short methods note on the systematic bias.

Q4. Where exactly is the *rmpA2* frameshift, and is it a single-lineage lesion or a conserved KpVP-1 pattern?

The paper never gives coordinates. Kleborate calls truncation at 60%. **Next steps:** extract *rmpA2* ORF from pVir-SCNJ1, align to pLVPK reference, localize the indel; repeat for all 30–40 KpVP-1 plasmids in NCBI; classify truncation position; correlate with hypermucoviscosity phenotype where wet-lab metadata allow. If systematic, this is a regulatory-tuning story rather than a random loss.

Q5. Seven years later — has ST29/K54/*bla*NDM-5/pNDM_MGR194-like plasmid spread internationally, or remained a Sichuan-restricted lineage?

The paper’s dataset stops March 2019. Our replication is 2019-vintage too. The pipeline (mash + abricate + Kleborate) is trivially re-runnable on a 2026 NCBI pull (likely 400–600+ IncX3 plasmids; ST29 WGS set likely doubled). **Next steps:** re-pull all *K. pneumoniae* genomes 2026-07; extract all ST29 isolates; re-run Parsnp+Gubbins+RAxML-NG with SCNJ1 as anchor; re-run Mash+NJ on the extended IncX3 set; map country/year of new close-relatives; estimate emergence rate of ST29-*bla*NDM-5 co-occurrence; test K54 enrichment. Deliverable: a “seven-year follow-up” rapid-communication note.

9 Reproducibility appendix

- **Paper PDF:** `paper.pdf`, sha256 204f058d324790ee989f89629bd54778c6712df94f51129a69b52a70c7e2790
- **Extraction:** `extraction/marker.md` (pdftotext -layout fallback; canonical Marker parse not resolved by paper sha256 in central SCOUT/OSTI corpus as of 2026-07-05); `extraction/nougat.mmd` placeholder (GPU parse pending).
- **Primary data:** NCBI RefSeq GCF_008320705.1 + comparators KF220657, NC_005249, NZ_CP031258; reference GCA_001630805; 59 ST29 assemblies + 231 IncX3 plasmids from GenBank (paper Tables S2 + S4).
- **Tools:** Kleborate 3.1.3, ABRicate 1.4.0, BLAST+ 2.16.0, Biopython 1.87, Parsnp 2.1.5, Gubbins 3.4.3, RAxML-NG 2.0.1, Mash 2.3, OrthoFinder 2.5.5, MAFFT 7.520, FastTree 2.1.11, Prodigal 2.6.3, NCBI Datasets 18.25.1.
- **Compute:** local host (Phase 1); chiatta00 128-core 2.1 TB JLSE Intel node (Phase 2).
- **Effort:** ~ 3 h wall-clock total across three sessions; ~ 75 CPU-hour equivalent for the phylogenies; ~ 55 agent tool-call steps.
- **LLM endpoints used:** Argo localhost:44497, Claude Opus family — free endpoints only; no paid API used.